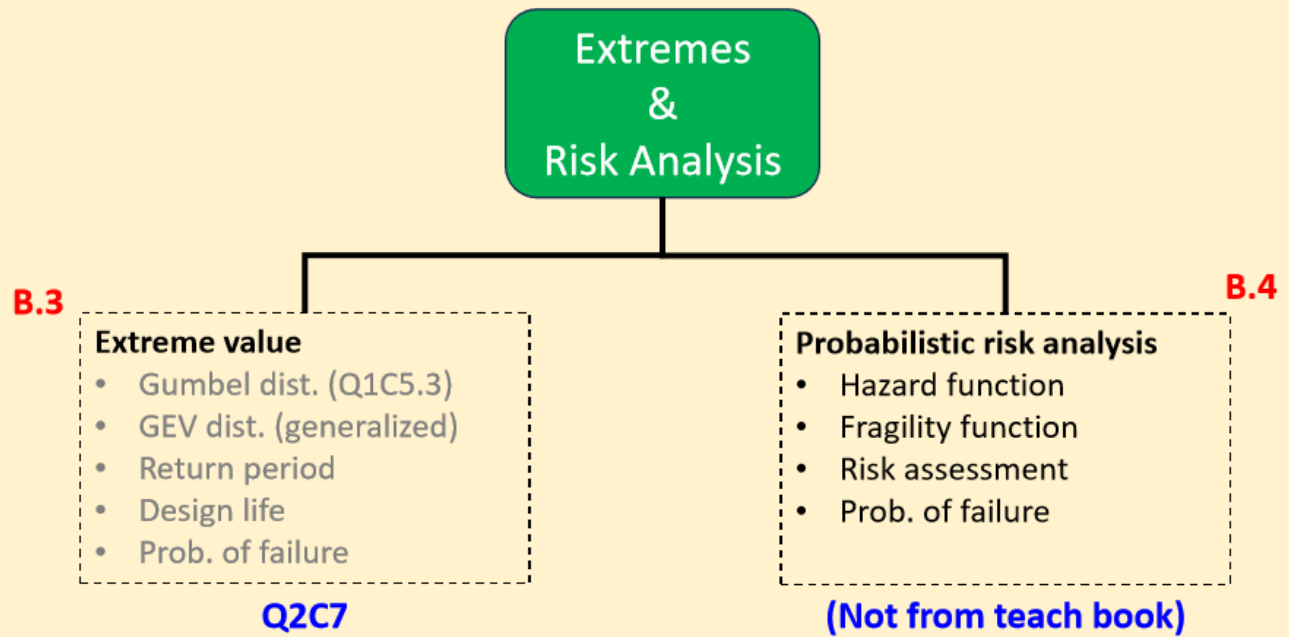


CV 5101
Modeling, Uncertainty, and
Data for Engineers
(July – Nov 2026)

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Module Overview



Flow

- Risk, reliability, resilience
- Typical probabilities
- Formalizing monkey attack
- Probabilistic monkey attack risk
- Probabilistic seismic risk

Risk, reliability, resilience

- p_f : Probability of Failure
- Risk: Expected loss
- Reliability: $P_r(\text{system performs its function without failure}) \approx 1 - p_f$
- Resilience: Ability of a system to withstand, recover, restore functionality after disruption.

Typical Probabilities

- Some **target probabilities** are:

- Locally dramatic but rather harmless in a wider area

e.g., building collapse

$\sim 10^{-4}$ for the lifetime of the facility

- More dramatic failure consequences, such as nuclear power plants

$\leq 10^{-5}$ /year core meltdown

$\leq 10^{-6}$ /year large leakage

- Drawing an ace of spade in a randomized deck of cards

$1/52 \approx 20 \times 10^{-3}$

- Rule of thumb,

probability of death/person/year $> 10^{-3} \Rightarrow$ Immediate Action

probability of death/person/year $< 10^{-6} \Rightarrow$ Event so unusual that

Nothing reasonable can be done.

Typical Probabilities

- Rule of thumb, **Probability of death/person/year**
 - $> 10^{-3} \Rightarrow$ Immediate Action
 - $< 10^{-6} \Rightarrow$ events so unusual that little can reasonably be done
 - Examples:
 - Worker in unregulated mine in early 20th century
often $> 10^{-2}$ / worker / year
 - Riding a motorbike without a helmet (in low-income countries)
 10^{-4}
 - Riding a motorbike with a helmet (in low-income countries)
 10^{-5}
 - Structural collapse of old and ill-maintained buildings
 - Commercial plane *1.2×10^{-8} / person-hour*
 - Car *0.7×10^{-8} / person-hour*
- Accounting for exposure*
- 24×10^{-6}*
 200×10^{-6}
- 6*

The R-S Reliability Problem

The R-S reliability problem

- Let R = capacity of a structure

S = demand on the structure

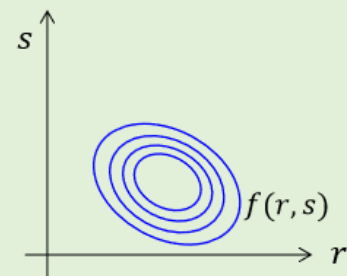
$f_{RS}(r, s)$ = joint PDF

Failure =

$p_f =$

$1 - p_f =$

$p_f =$



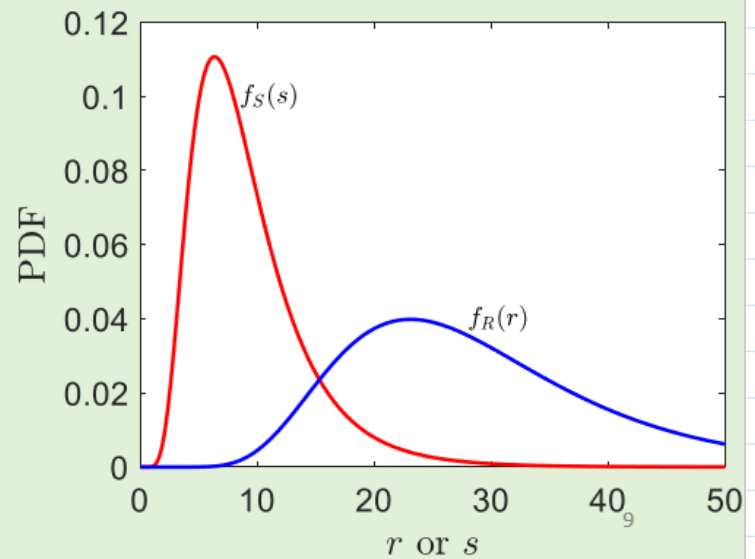
The R-S reliability problem

- Solution by conditioning on S

$$f_{RS}(r, s) =$$

$$p_f = \Pr(R \leq S) =$$

If R and S are statistically independent,



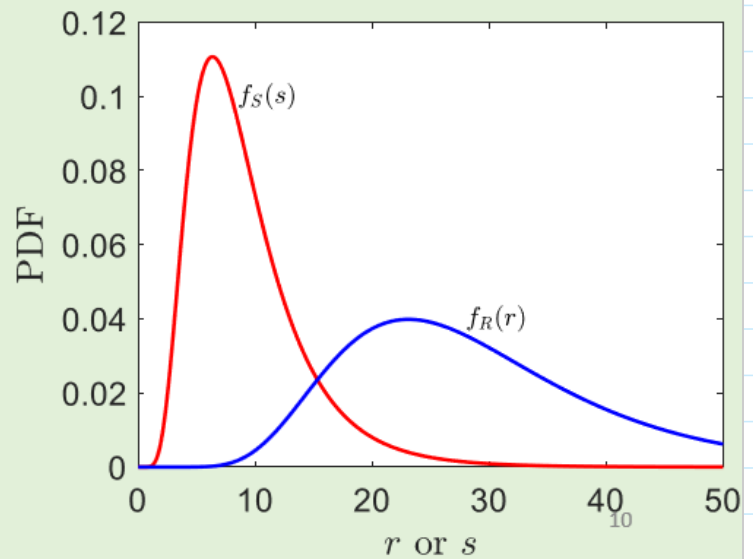
The R-S reliability problem

- Solution by conditioning on R

$$f_{RS}(r, s) =$$

$$p_f = \Pr(R \leq S) =$$

If R and S are statistically independent,



Probabilistic Risk Assessment



Risk from monkey attack on campus

- The risk from the monkey hazard involves:

- Probability of Monkey Attack

- Location, the human behavior,

• Extents: CHARGING → SCRATCHING → BITING

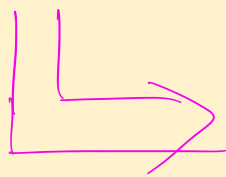
↓
MAULING

MONKEY ATTACK SCALE, MAS

- Rate of Monkey Attack: HOW OFTEN? HAZARD RATE

- Damage from Monkey Attack:

hospitalization duration: ZERO → HOURS → DAYS → WEEKS +



Risk of Monkey Attack.

Probabilistic monkey risk assessment

- Three components

Given Location & Person (Z & P)

- Monkey Hazard: $H_{MAS} := \Pr(MAS = mas | Z)$

- Person's Fragility: $F_{Dam, P} := \Pr(DAM = dam | MAS = mas)$

- Consequence: $C_{DT} := \Pr(DT = dt | DAM = dam)$

downtime $\leftarrow$

$$\lambda_{DT}(DT = dt) = \int \int \frac{p(DT | dam) \cdot p(DM | mas)}{P(MAS | Z)} dMAS dDM$$

Probabilistic seismic risk assessment

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Questions, comments,
or concerns?